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IEOR 221 – Homework 6

Module 7: Options – BSM Equations

Problem 1. Probability of Negative Cash Position (Hull 14.12)

Let the company's cash position X_t follow

$$dX = \mu\, dt + \sigma\, dW,$$

where the drift rate per quarter is $\mu = 0.5$ and the variance rate is $\sigma^2 = 4.0$, so $\sigma = 2.0$. Over one year ($T = 4$ quarters),

$$X_T \sim \mathcal{N}(X_0 + \mu T,\, \sigma^2 T) = \mathcal{N}(X_0 + 2,\, 16).$$

We require $P(X_T < 0) = 0.05$. Then

$$P\left(Z < \frac{0 - (X_0 + 2)}{4}\right) = 0.05, \qquad \frac{-(X_0 + 2)}{4} = -1.6449.$$

Solving gives $X_0 + 2 = 4(1.6449)$, hence

$$X_0 = 4.5796 \approx 4.58.$$

The firm must start with at least 4.58 million dollars of cash to have less than a 5% chance of a negative balance after one year.

Problem 2. Comparison of Stochastic Models (Hull 14.17)

The reference model is

$$\Delta S = \mu S\, \Delta t + \sigma S\, \Delta W.$$

(i) $\Delta S = \mu\, \Delta t + \sigma\, \Delta W$: both expected change and volatility are constant in dollar terms, independent of price. Price could become negative and relative changes vary with level.
(ii) $\Delta S = S\, \Delta t + \sigma\, \Delta W$: the drift scales with S but volatility remains absolute, which implies the same dollar volatility for expensive and cheap stocks.
(iii) $\Delta S = \mu\, \Delta t + \sigma S\, \Delta W$: the volatility term scales properly with price but the drift term does not, so expected percentage growth is not constant.

In contrast, the geometric Brownian motion

$$\Delta S = \mu S\,\Delta t + \sigma S\,\Delta W$$

ensures both drift and diffusion are proportional to S. This keeps the percentage return approximately normal with mean $\mu\Delta t$ and variance $\sigma^2\Delta t$, matching the empirical property that stock returns, rather than absolute price changes, have roughly constant variance.

Problem 3. Process for S^n (Hull 14.19)

Given $dS = \mu S\,dt + \sigma S\,dW$, let $Y = S^n$. Applying Itô's lemma,

$$dY = nS^{n-1}dS + \tfrac{1}{2}n(n-1)S^{n-2}(dS)^2.$$

Substituting dS,

$$dY = nS^{n-1}(\mu S\,dt + \sigma S\,dW) + \tfrac{1}{2}n(n-1)\sigma^2 S^n\,dt.$$

Hence

$$dY = S^n\left[(n\mu + \tfrac{1}{2}n(n-1)\sigma^2)\,dt + n\sigma\,dW\right].$$

Therefore S^n also follows a geometric Brownian motion with drift $n\mu + \frac{1}{2}n(n-1)\sigma^2$ and volatility $n\sigma$.

Problem 4. One-Day Stock Price Distribution (Hull 14.22, 250 trading days)

Parameters:

$$S_0 = 50, \qquad \mu = 0.16, \qquad \sigma = 0.30, \qquad \Delta t = \frac{1}{250}.$$

Under GBM,

$$E[S_{t+\Delta t}] = S_0 e^{\mu\Delta t}, \qquad \mathrm{Var}(S_{t+\Delta t}) = S_0^2 e^{2\mu\Delta t}\left(e^{\sigma^2\Delta t} - 1\right).$$

(a) Expected price:

$$E[S_{t+\Delta t}] = 50\,e^{0.16/250} = 50.0320.$$

(b) Standard deviation:

$$\mathrm{SD} = 50\,e^{0.16/250}\sqrt{e^{0.30^2/250} - 1} = 0.9494.$$

(c) 95% confidence interval (normal approximation):

$$E[S] \pm 1.96\,\mathrm{SD} = 50.0320 \pm 1.96 \times 0.9494 \Rightarrow [\,48.1712,\ 51.8928\,].$$